In [1]:
```python
import pandas as pd
from sklearn.metrics import confusion_matrix, classification_report, accuracy_
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
```

In [ ]:
```python
df = pd.read_csv('datasets/Iris.csv')
df
```

Out[ ]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Speci
0	1	5.1	3.5	1.4	0.2	Ir setc
1	2	4.9	3.0	1.4	0.2	Ir setc
2	3	4.7	3.2	1.3	0.2	Ir setc
3	4	4.6	3.1	1.5	0.2	Ir setc
4	5	5.0	3.6	1.4	0.2	Ir setc
...	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	Ir virgini
146	147	6.3	2.5	5.0	1.9	Ir virgini
147	148	6.5	3.0	5.2	2.0	Ir virgini
148	149	6.2	3.4	5.4	2.3	Ir virgini
149	150	5.9	3.0	5.1	1.8	Ir virgini

150 rows × 6 columns

In [3]:
```python
df.isnull().sum()
```

Out[3]:
```
Id               0
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

In [4]:
```python
label_encoder = LabelEncoder()
df['Species'] = label_encoder.fit_transform(df['Species'])
```

```
df
```

Out[4]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Speci
0	1	5.1	3.5	1.4	0.2	
1	2	4.9	3.0	1.4	0.2	
2	3	4.7	3.2	1.3	0.2	
3	4	4.6	3.1	1.5	0.2	
4	5	5.0	3.6	1.4	0.2	
...	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	
146	147	6.3	2.5	5.0	1.9	
147	148	6.5	3.0	5.2	2.0	
148	149	6.2	3.4	5.4	2.3	
149	150	5.9	3.0	5.1	1.8	

150 rows × 6 columns

In [5]:

```
x = df.drop('Species', axis=1)
x
```

Out[5]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	1	5.1	3.5	1.4	0.2
1	2	4.9	3.0	1.4	0.2
2	3	4.7	3.2	1.3	0.2
3	4	4.6	3.1	1.5	0.2
4	5	5.0	3.6	1.4	0.2
...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3
146	147	6.3	2.5	5.0	1.9
147	148	6.5	3.0	5.2	2.0
148	149	6.2	3.4	5.4	2.3
149	150	5.9	3.0	5.1	1.8

150 rows × 5 columns

In [6]:

```
y = df.Species
y
```

Out[6]:
```
0      0
1      0
2      0
3      0
4      0
      ..
145    2
146    2
147    2
148    2
149    2
Name: Species, Length: 150, dtype: int64
```

In [7]:
```
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, randc
gaussian = GaussianNB()
gaussian.fit(x_train, y_train)
```

Out[7]:

▾ GaussianNB

Parameters

	priors	None
	var_smoothing	1e-09

In [8]:
```
y_pred = gaussian.predict(x_test)
```

In [9]:
```
matrix = confusion_matrix(y_test, y_pred)
matrix
```

Out[9]:
```
array([[11,  0,  0],
       [ 0, 13,  0],
       [ 0,  0,  6]])
```

In [10]:
```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        11
           1       1.00      1.00      1.00        13
           2       1.00      1.00      1.00         6

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

In [11]:
```
accuracy = accuracy_score(y_test, y_pred)
accuracy
```

Out[11]:
```
1.0
```

In [12]:
```
class_names = label_encoder.classes_
```

```python
total = len(y_test)
n_classes = len(class_names)

for i in range(n_classes):
    TP = matrix[i][i]
    FP = sum(matrix[j][i] for j in range(n_classes)) - TP
    FN = sum(matrix[i]) - TP
    TN = total - TP - FP - FN
    precision = TP / (TP + FP) if (TP + FP) > 0 else 0
    recall = TP / (TP + FN) if (TP + FN) > 0 else 0

    print(f'Class {i} ({class_names[i]}):')
    print(f'  True Positive  (TP): {TP}')
    print(f'  True Negative  (TN): {TN}')
    print(f'  False Positive (FP): {FP}')
    print(f'  False Negative (FN): {FN}')
    print(f'  Precision: {precision}')
    print(f'  Recall:    {recall}')
    print()

print(f'Overall Accuracy:   {accuracy}')
print(f'Overall Error Rate: {1 - accuracy}')
```

```
Class 0 (Iris-setosa):
  True Positive  (TP): 11
  True Negative  (TN): 19
  False Positive (FP): 0
  False Negative (FN): 0
  Precision: 1.0
  Recall:    1.0

Class 1 (Iris-versicolor):
  True Positive  (TP): 13
  True Negative  (TN): 17
  False Positive (FP): 0
  False Negative (FN): 0
  Precision: 1.0
  Recall:    1.0

Class 2 (Iris-virginica):
  True Positive  (TP): 6
  True Negative  (TN): 24
  False Positive (FP): 0
  False Negative (FN): 0
  Precision: 1.0
  Recall:    1.0

Overall Accuracy:   1.0
Overall Error Rate: 0.0
```